

Relay

An on-device, voice-controlled, screen-reading accessibility agent

Snapdragon AI Lab Build & Present Challenge

Point-and-click assumes hands and eyesight most people don't have

- People with limited hand mobility or low vision can't reliably operate a Windows PC through a mouse and a screen alone.
- Existing screen readers describe a screen; they don't act on it. Existing automation agents act, but weren't built for this population's safety needs.
- The gap: a voice-first agent that both understands the screen and acts on it — with a person narrating every step.

"Open my email and read me the new ones" — that's the whole interaction

- Voice in, spoken result out. No pointing, no reading required.
- Deliberately not a general autonomous computer-use agent — scoped assistive technology.
- A person narrates every step, and nothing consequential happens without a spoken "yes." That's a feature of the design, not a limitation.

Screen understanding is tree-first, vision-second — not the other way around

Voice command

Whisper (speech-to-text)

Screen understanding — UI Automation tree first, vision model on fallback

Small local reasoning model decides the next action

Guardrail check — ALLOW / CONFIRM (spoken yes) / BLOCK

Action executes against the real UI element

MeloTTS speaks the result

Relay is the only one with a deterministic, code-level safety gate

Project	Safety mechanism	Enforcement
Deft	None found anywhere in either backing repo	N/A — unconditional dispatch
Orbit	None — but never takes actions at all	N/A
ScreenSaathi	Keyword list downgrades risky plans to guide-only	Code-level, deterministic
Sally	A prompt instruction asking the model nicely	Model-enforced only — no code backstop
Relay	Two-tier ALLOW / CONFIRM / BLOCK + spoken yes + audit log	Code-level, deterministic, pre-execution

Sourced from reading each project's real, public source code — not a secondhand summary. See COMPETITIVE_RESEARCH.md.

Every non-NPU piece of the pipeline has been run for real, on this machine

- Whisper speech-to-text: real transcription of a known test clip, 3 independent ways.
- MeloTTS-EN: real synthesis, real audio, played back through real speakers.
- The full agent.py orchestration loop: a real mic recording → real transcription → real UI Automation tree read → real guardrail gate → real action → real speech, proven end to end against a live window.
- Chromium accessibility gap (Edge/Outlook can't be read via UI Automation): found, verified, and fixed for Edge via Chrome DevTools Protocol — demonstrated live against a real Wikipedia page.
- The one honest gap: LocalReasoner itself needs Snapdragon's on-device NPU (GenieX), which this dev machine doesn't have.

Whisper-base, compiled and profiled on a real Snapdragon X Elite

~49.1 ms

encoder

~3.7 ms

decoder

A real compile + profile job on Qualcomm AI Hub's hosted device farm — not a simulation, and an order of magnitude faster than this laptop's CPU. Precisely scoped: this is AI Hub's compile/profile path, not yet a full GenieX runtime measurement, which needs physical device access.

The end user in the moment of use, and whoever configures the guardrails

- End-user runtime UI: 7 explicit states (listening, transcribing, deciding, confirming, acting, blocked, error) — WCAG AAA contrast (7:1), verified by computed contrast ratios, not eyeballed.
- The confirmation screen is the single highest-stakes moment in the product: distinct shape and color, never color alone, spoken aloud as well as shown.
- Guardrail-admin surface: reads guardrails.py's real CONFIRM/BLOCK patterns directly
- can't drift out of sync with a hand-maintained copy.

What's real, what's next

- Open: Outlook's Chromium accessibility gap — same limitation as Edge, but it's a desktop app CDP can't attach to. Still unsolved, documented as such.
- Open: full GenieX-runtime, on-device latency for the reasoning model — needs physical Snapdragon hardware, not yet measured.
- Done: 18 adversarial guardrail tests across all 3 target apps, a fixed task-list harness (100% pass rate), and short-term reasoning memory (last 3 turns).
- This submission was built without physical Snapdragon hardware — every claim here is labeled by exactly how it was verified.

Thank you

github.com/kartik1pandey/relay-accessibility-agent

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